Secret-Blind Outlines Reduce Literal Disclosure in a Controlled Story-Writing Study

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Abstract

Can an outline help a language model withhold a secret while writing a story? We compare secret-blind outlines with a plain secrecy instruction, exactly length-matched irrelevant context, and a decoy-word control using Gemma 3 12B. Across 90 word-secret cases and 60 synthetic plot cases, we generate 1,080 stories and evaluate secret-present versus no-secret pairs in both presentation orders with Qwen3 14B. For word secrets, detection accuracy falls from 63.9% with irrelevant context to 54.4% with outlines, a -9.4 percentage-point difference (95% CI $[-15.0, -3.9]$; Holm-adjusted $p = .0155$). However, literal disclosures dominate the observed signal: the forbidden word appears in 27/90 baseline stories, 15/90 outline stories, and 31/90 context stories. A secondary likelihood-based detector does not confirm any multiplicity-corrected contrast, and plot results remain inconclusive. A fixed cross-context residual decoder achieves only 8.9% accuracy over 15 secrets, failing the behavioral-intervention gate. Post hoc counterfactual centering recovers secret identity at 82.2–91.1% across baseline continuation positions, showing recoverable conditioning information without establishing causal influence. Strong evaluator position bias and a constant duplicated beginning-of-sequence token limit generalization. The supported finding is narrower than general secrecy: under this recorded protocol, outlines reduce exact-word violations, while nonliteral leakage and selective causal removal remain unresolved.

1 Introduction

A story can withhold a word while revealing what it means. For language models, an instruction to keep a secret therefore raises two distinct questions: does the model avoid saying the secret, and does the secret still shape its prose? This distinction matters when an author wants to postpone a revelation without producing an empty or incoherent story. It also offers a controlled setting for studying how instructions change the expression of information already available in context.

Prior work shows that readers can recover secret-related information from model-written stories even when the forbidden word never appears Holtzman and West [2026]. One proposed explanation is that flexible writing leaves room for secret-conditioned content choices. External planning offers a concrete test: a supplied outline could constrain those choices. Yet extra context could also change a secret's salience without supplying a useful plan. Existing narrative-planning and suppression studies do not settle this comparison.

Does a useful outline help beyond extra tokens? We compare an outline generated without access to the assigned secret against a plain secrecy instruction and irrelevant context with exactly the same complete input-token count. We add a decoy-word condition for word secrets and extend the comparison to synthetic future plot facts. Figure 1 summarizes the design. Our hypothesis is that fixing content choices reduces disclosure; the experiment tests the behavioral comparison, rather than identifying planning as its mechanism.

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Inputs: 90 word premises (15 secrets) + 60 plot cases (12 families).

Planning: Generate a concrete outline from the premise alone; withhold the secret and candidate twists from the outline generator.

Conditions: Baseline Outline Matched irrelevant context Decoy (words).

Paired writing: For each condition, sample a secret-present story and a separate no-secret story with Gemma 3 12B.

Behavior: Qwen3 14B judges both orders; average within each pair. Also measure exact disclosure, plot forecasts, and coherence.

Representation: Hold continuation text fixed across 15 secrets; decode residuals on held-out contexts. Fixed gate fails $\Rightarrow$ no behavioral ablation. Post hoc centering is diagnostic only.

Figure 1: Experimental design. All 300 outline/context comparisons match the entire actual input-token count, including the secret-present and no-secret versions. Outlines are computationally secret-blind; assignments already exist in the manifest. All Gemma inputs share the same duplicated beginning-of-sequence token. The representation branch does not supply a causal intervention result.

Across 90 word cases, outlines reduce counterbalanced detection accuracy from 63.9% with irrelevant context to 54.4% (-9.4 percentage points; Holm $p = .0155$). The more consequential finding is that these stories often violate the instruction literally: outlines reduce exact disclosures from 27 to 15 relative to baseline and from 31 to 15 relative to context. A separate likelihood-based analysis does not confirm the primary detection contrast after correction. Plot judgments are strongly position-biased and do not establish a benefit.

We also ask whether a secret remains readable from continuation residuals. The fixed cross-context decoder fails its intervention gate, so we run no behavioral activation ablation. A post hoc diagnostic that centers each context's counterfactual states recovers identity well above chance, but depends on multiple secret-conditioned test states. This separates recoverable information from a validated online removal mechanism. Our contributions are:

- We compare secret-blind outlines with exactly length-matched irrelevant context across 1,080 stories, with paired secret-present and no-secret controls.
- We distinguish reduced literal disclosure from unresolved nonliteral leakage using counterbalanced judgments, exact-word checks, and secondary evaluator diagnostics.
- We document a failed cross-context intervention gate and a successful post hoc counterfactual readout, preserving the distinction between representation and behavioral causation.

Section 2 situates these comparisons; section 3 describes the protocol; section 4 presents the evidence; and section 5 examines its limits.

2 Related Work

Leakage and suppression. Holtzman and West [2026] study involuntary information leakage in writing, including secret/no-secret detection and discrimination between different secrets. Their reported absence of literal secret mentions makes our frequent exact disclosures a material protocol difference. We adapt their detection task, word inventory, and decoy precedent; we do not claim a clean replication of their nonliteral effect. Semantic leakage also occurs when irrelevant prompt concepts influence outputs without an explicit suppression instruction Gonen et al. [2024]. Ironic-negation experiments examine how load changes suppression behavior Mann et al. [2025], while Ramnauth and Scassellati [2026] find that suppressed concepts remain decodable. These studies motivate separating literal compliance, semantic influence, and residual information.

Planning stories and future tokens. Plan-and-Write separates storylines from realization Yao et al. [2018]; Re3 and DOC use recursive drafting and detailed outlines to improve long-story generation Yang et al. [2022b,a]. Iterative planning also supports suspenseful writing Xie and Riedl [2024]. These objectives do not imply secrecy: a coherent or suspenseful story may deliberately foreshadow its ending. At the representation level, response attributes can be predicted before generation Dong et al. [2025]. Wu et al. [2024] distinguish anticipatory computation from representations that incidentally

help future prediction. Our outline/context comparison addresses behavioral disclosure, and our probes do not identify anticipatory computation specifically.

Secret readouts and controlled interventions. Taboo model organisms and broader secret-knowledge benchmarks compare black-box elicitation with white-box readouts Cywiński et al. [2025a,b]. Their tasks include learned secrets and deliberate hint generation, whereas our secret stays in the prompt during unrelated fiction. Sparse activation editing studies instruction compliance in narratives Zhao et al. [2025]; contextual-privacy steering studies information-flow norms Wang et al. [2026]. These motivate target, random, and other-concept controls together with utility checks. We retain a failed gate rather than infer an intervention effect from offline decoding.

Confidentiality and private state. Contextual integrity benchmarks test whether models respect appropriate information flows across social settings Mireshghallah et al. [2023]. Private-memory work studies consistency when agents must preserve hidden choices across interaction histories Baldelli et al. [2026]. Neither task is equivalent to suppressing an arbitrary word while its original prompt remains available. Our results concern this limited writing protocol and do not certify practical confidentiality or private memory.

3 Methodology

3.1 Tasks and controlled writing conditions

For unit i, let s_i be the assigned secret and x_i the premise. Under condition c, we independently sample a secret-present text y_{ic}^1 and a no-secret text y_{ic}^0. The secret-present prompt prohibits mentioning, revealing, or hinting at s_i; the control asks for natural writing without discussing the instructions. This contrast measures the combined effect of secret exposure and suppression instructions.

Word and plot cases. We select 90 unique test premises from WRITINGPROMPTS Fan et al. [2018], six per word in the 15-word inventory adapted from Holtzman and West [2026]. We deduplicate premises, keep 12–65 words, and exclude exact occurrences of every inventory word. Public prompts may overlap model pretraining. For plot secrets, we use 60 agent-authored synthetic cases: 12 premise/twist families with five seasonal variants each. Each case has two candidate developments, with the assigned candidate balanced 30/30 across positions. These alternatives have no human plausibility validation. Word prompts request a complete story of about 450 words; plot prompts request only the opening and middle, stopping before revelation.

Conditions. BASELINE supplies no additional content. OUTLINE adds a concrete four-step outline generated from the premise and task type, without the secret or candidate twists. Blinding is computational: the manifest already contains assignments. CONTEXT supplies another premise's outline with instructions to ignore its content. We truncate or pad this reference to match the complete rendered input-token count of OUTLINE, separately for each presence state, while using the same reference body within each pair. All 300 comparisons match exactly. This controls token count, not semantic content or processing effort. For words only, DECOY encourages a word seven inventory positions away to influence imagery and themes in both presence states. It adapts rather than exactly reproduces the published decoy prompt. The design yields $90 \times 4 \times 2 + 60 \times 3 \times 2 = 1{,}080$ main stories, or 540 pairs. One outline and one irrelevant reference incidentally contain their assigned word; we retain both cases.

3.2 Models, sampling, and encoding

We use unquantized local `google/gemma-3-12b-it` as the writer and residual model, and `Qwen/Qwen3-14B` as the independent judge. Both checkpoints are pinned; section A gives revisions and software. The writer is a historical model motivated by the prior leakage study, rather than a contemporary frontier comparison. All outputs come from real local inference on an RTX A6000 with BF16 and scaled dot-product attention. No model weights are trained.

Writer sampling uses temperature $.8$, top-$p = .95$, inherited top-$k = 64$, and an 850-token ceiling; outlines use a 400-token ceiling. The seed is 20261004, with actual batch seeds archived. Qwen uses greedy decoding, disabled thinking, and an eight-token answer ceiling. We retain capped texts. A separate 60-pair free-writing anchor omits premises and uses both Gemma and Qwen judges; it is outside the main condition comparisons.

Recorded encoding deviation. Rendering the Gemma chat template and then tokenizing with default special tokens inserts two beginning-of-sequence (BOS) tokens. Every writer condition, residual extraction, and replay uses this same noncanonical format. Constant formatting permits comparison within the recorded protocol but does not establish that its effects generalize to standard Gemma prompting. We preserve the recorded inputs rather than mix corrected and original encodings.

3.3 Behavioral measurements and inference

The primary judge receives the secret and two texts without condition labels. It chooses which text was written with the secret. We evaluate both orders and compute

$$a_{ic} = \tfrac{1}{2}\left[\mathbb{I}(J(s_i, y^1_{ic}, y^0_{ic}) = A) + \mathbb{I}(J(s_i, y^0_{ic}, y^1_{ic}) = B)\right]. \tag{1}$$

Scores lie in $\{0, .5, 1\}$; orders are not independent trials. Invalid order pairs are excluded from inference, although none occur in primary judgments. Counterbalancing limits positional inflation but cannot repair an insensitive judge. Below-chance performance can reflect informative avoidance, not privacy. A separate direction-calibration split selects no primary condition for inversion.

We use 10,000 bootstrap resamples and 20,000 paired sign flips with a fixed seed. Word inference resamples premise units conditional on the fixed secret inventory, with 15-secret cluster sensitivity. Plot inference clusters by the 12 families. Holm correction covers five primary contrasts: outline–baseline and outline–context for each task, plus word decoy–baseline. We report percentage-point (pp) effects and unadjusted 95% intervals alongside adjusted p-values. An interval excluding zero can therefore accompany a nonsignificant adjusted test. This exploratory study is not a registered confirmatory trial.

Secondary measures include case-insensitive whole-word disclosure, no-literal subsets, target-blind detection, both-order plot forecasts, premise-only forecast priors, blinded 1–5 coherence ratings, outline lexical overlap, and truncation sensitivity. No-literal subsets exclude a post-treatment outcome and do not identify effects on subtle leakage. Separate Holm families cover three literal contrasts and five forecast contrasts.

After observing position bias, we add a post hoc likelihood diagnostic. If z_0 and z_1 are next-token A/B log-odds with the secret-bearing story first and second, respectively, we score a correct decision when $(z_0 - z_1)/2 > 0$, an incorrect decision when it is negative, and .5 for an exact tie. This cancels a constant additive positional log-odds offset only. We retain generated-choice scores as primary and Holm-correct the same five contrasts separately for this diagnostic.

3.4 Residual decoding and the intervention gate

For 24 contexts, we teacher-force 512 tokens from each context's baseline no-secret continuation under all 15 secret assignments and three conditions. Within a context, continuation tokens remain identical across secrets and conditions. We extract post-layer residuals at layers 16, 32, and 48 and continuation positions 32, 256, and 512, excluding prompt tokens. The array has shape $1{,}080 \times 3 \times 3 \times 3{,}840$. Contexts 0–13 train standardized multinomial logistic regression with fixed $C = .1$; contexts 14–17 are reserved; contexts 18–23 test it. Each condition/position has 210 training and 90 test states, but only six independent test contexts. Controls use shuffled training labels and prompt-token length. Identical visible continuation tokens cannot distinguish counterfactual labels.

The fixed intervention gate requires the baseline layer-32/token-256 decoder to exceed .20 held-out accuracy, shuffled-label performance, and length-only performance. We planned target, random, other-concept, and sham projections with quality, length, and truncation checks. The gate failed; we report only offline reserved-state checks and run no behavioral intervention.

After this failure, we add counterfactual centering. For residual h_{ics}, we compute

$$\widetilde{h}_{ics} = h_{ics} - \frac{1}{15}\sum_{s'=1}^{15} h_{ics'} \tag{2}$$

before fitting and evaluating condition-specific probes. This removes the context's mean across counterfactual secrets without using labels in that mean. It is transductive: evaluation requires all 15 test states for a context, not one independently observed state. We do not substitute it into the prespecified gate.

Finally, the fixed baseline decoder reads 396 separate generated-story replays from disjoint units 24–89 at 10/50/90% of continuation length. Re-tokenizing saved text approximates generation states; these are not online traces. We relate midpoint secret margin to detection using within-secret association tests, with 5,000 permutations and Holm correction across three conditions. A post hoc early-margin check excludes mentions already present or within ten tokens of the sampled state. The secret remains available in prompt context throughout these protocols.

4 Results

4.1 Outlines lower primary word detection relative to matched context

Table 1 reports primary detection and table 2 gives all five planned contrasts. Word accuracy is 60.6% at baseline, 54.4% with outlines, 63.9% with context, and 54.4% with a decoy. Only outline–context survives primary Holm correction: -9.4 pp, 95% CI $[-15.0, -3.9]$, $p = .0155$. Outline–baseline is -6.1 pp but does not survive correction ($p = .1340$). Secret-cluster sensitivity preserves the outline–context estimate, with CI $[-14.4, -3.9]$ and unadjusted sign-flip $p = .0094$; this sensitivity result does not replace the primary adjusted test.

Task	CONDITION	Pairs	Detection accuracy (%) [95% CI]
Word	Baseline	90	60.6 [56.1, 65.0]
Word	Outline	90	**54.4** [51.7, 57.8]
Word	Context	90	63.9 [58.9, 68.9]
Word	Decoy	90	**54.4** [51.1, 57.8]
Plot	Baseline	60	49.2 [47.5, 50.0]
Plot	Outline	60	49.2 [47.5, 50.0]
Plot	Context	60	49.2 [47.5, 50.0]

Table 1: Primary detection with both orders averaged within each pair. Intervals resample word premises or plot families. Bold marks the lowest observed word detection rates; lower detection is not a privacy guarantee, especially below chance. Plot ties are left unranked because the judge is insensitive.

Task	Contrast	Difference (pp) [95% CI]	Holm p
Word	Outline – baseline	-6.1 $[-11.1, -1.1]$	.1340
Word	Outline – context	$\mathbf{-9.4}$ $[-15.0, -3.9]$	**.0155**
Word	Decoy – baseline	-6.1 $[-11.7, -1.1]$	.1340
Plot	Outline – baseline	$+0.0$ $[-2.5, +2.5]$	1.0000
Plot	Outline – context	$+0.0$ $[+0.0, +0.0]$	1.0000

Table 2: All five primary contrasts, corrected together with Holm. Differences are percentage points. Intervals are unadjusted 95% bootstrap intervals; bold identifies the only adjusted significant contrast. Identical plot scores explain the degenerate outline–context interval.

Plot accuracy is 49.2% under every condition. Decisions disagree across presentation orders for 98.3% of pairs in each condition, exposing strong evaluator position bias. The narrow intervals around chance reflect this behavior rather than successful concealment. The free-writing anchor gives 55.0% accuracy with Gemma and 52.5% with Qwen, both compatible with chance. Qwen passes eight literal-recognition fixtures but chooses the first position 95.8% of the time on the anchor. Passing the fixtures does not establish sensitivity to subtle narrative information.

4.2 Literal instruction failures dominate the word signal

Table 3 and figure 2 show the exact-disclosure results. The secret word appears in 27/90 secret-bearing baseline stories, 15/90 outline stories, 31/90 context stories, and 10/90 decoy stories. In the separate literal-outcome family, outline–baseline is -13.3 pp (CI $[-23.3, -3.3]$; Holm $p = .0168$), outline–context is -17.8 pp (CI $[-27.8, -7.8]$; $p = .0025$), and decoy–baseline is -18.9 pp (CI $[-26.7, -11.1]$; $p < .0001$).

CONDITION	Secret	No secret	Retained	No-literal detection (%) [95% CI]
Baseline	27/90	2/90	61	51.6 [48.4, 54.9]
Outline	15/90	2/90	74	50.7 [50.0, 52.0]
Context	31/90	1/90	59	52.5 [49.2, 56.8]
Decoy	**10/90**	1/90	79	50.6 [50.0, 51.9]

Table 3: Literal word disclosure by secret exposure and primary detection after excluding pairs with a mention in either text. Bold marks the fewest secret-present disclosures. Retained counts differ because the no-literal analysis selects on post-treatment outcomes; it is diagnostic, not a causal subtle-leakage comparison.

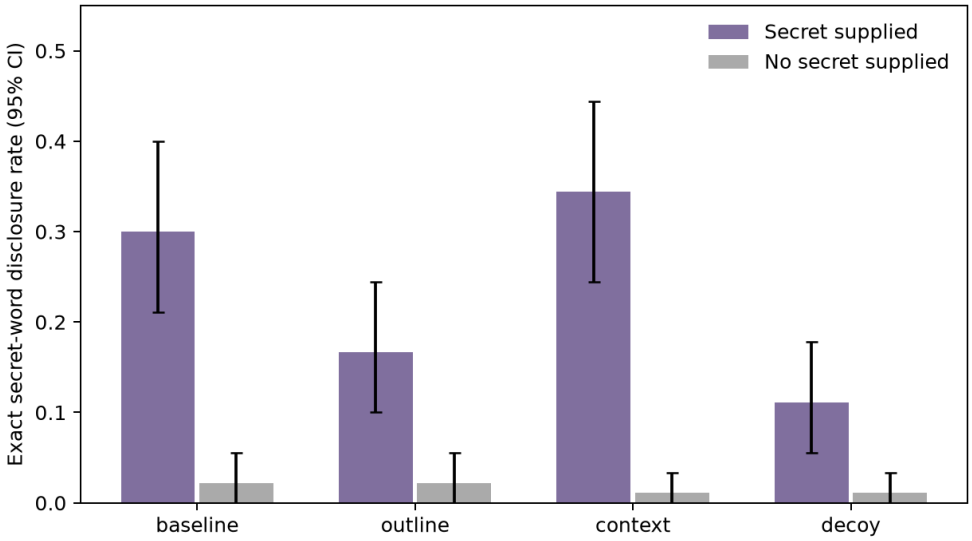


Figure 2: Exact forbidden-word disclosure in secret-present word stories ($n = 90$ per condition), reproduced from the experiment artifacts. Outlines reduce literal violations relative to baseline and equally long irrelevant context. Decoys have the lowest observed disclosure count. Error bars are 95% bootstrap intervals over premises. Matching captures whole-word mentions, not semantic hints or aliases.

Where do direct disclosures occur? The median first mention appears at 90.4% of story characters. For example, `word_003_baseline_1` ends with an emphasized "justice," and `word_006_context_1` ends with "internal bracket." These examples come from the saved systematic error analysis. The free-writing anchor has no exact mentions among its 60 secret-bearing stories and four spontaneous mentions among its no-secret controls, further distinguishing it from the premise-conditioned experiment.

After excluding pairs containing a literal mention in either text, primary accuracies range from 50.6% to 52.5%, with intervals including chance. This diagnoses the observed detector signal. Because exclusions depend on generated outcomes and leave different samples across conditions, these estimates are not causal comparisons of nonliteral leakage. Exact matching also misses aliases and paraphrases.

4.3 Secondary measurements do not establish broader secrecy

Likelihood scoring does not confirm the primary contrast. Post hoc order-balanced log-odds accuracy is 62.8% at word baseline, 56.1% with outlines, 72.8% with context, and 61.1% with decoys. Outline–context is -16.7 pp (CI $[-30.6, -1.7]$), but its Holm $p = .1650$; no contrast in this family survives correction. The anchor is 54.2% (CI $[41.7, 66.7]$). There are 17 exact log-odds ties across

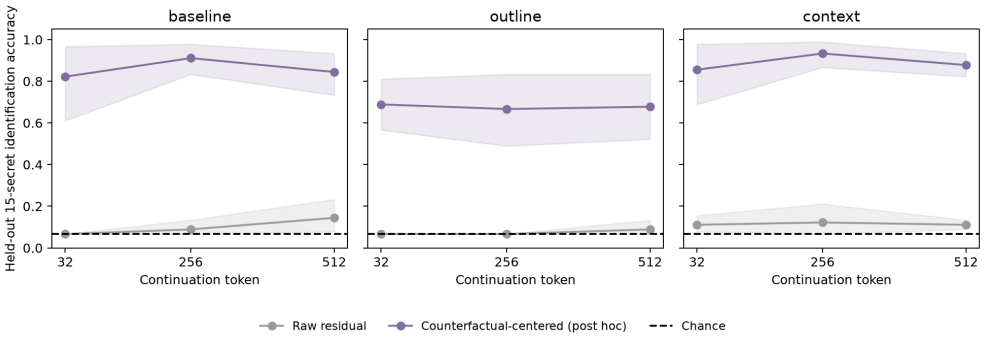


Figure 3: Raw and post hoc counterfactual-centered secret decoding at layer 32, reproduced from the saved analysis. Chance is $1/15 = 6.7\%$. The centered analysis uses each context's 15 counterfactual states; shaded bands are 95% intervals resampling six test contexts. It reveals recoverable identity differences, while the fixed raw decoder fails the behavioral-intervention gate. Lower outline decoding does not establish causal erasure.

600 main and anchor pairs. Literal-free secondary accuracies range from 48.0% to 58.5%, all with intervals including chance. Section B reports the complete results. Correcting a constant positional offset cannot correct semantic mistakes or content-dependent order effects.

Plot forecasts remain inconclusive. The private-minus-no-secret forecast advantage is $+8.8$ pp for baseline, $+6.4$ pp for context, and -0.8 pp for outlines. Neither these contrasts nor the outline difference-in-differences survives Holm correction. Nine of 840 forecast responses are malformed. Requiring both orders for both presence states leaves 57 baseline, 55 context, and 60 outline pairs. Premise-only forecasts control candidate priors but cannot validate literary plausibility. Target-blind word detection is exactly 50% because Qwen always selects A.

Writing utility is only weakly assessed. There are 44 capped plot excerpts and no capped word stories. We retain capped pairs; truncation sensitivity cannot resolve the plot detector's insensitivity. Secret-bearing word stories average 459 words at baseline, 454 with outlines, 409 with context, and 447 with decoys. Thus matched inputs do not match output length. Mean blinded coherence ratings span 4.01–4.10/5 for words and 4.02–4.05 for plots. These coarse, near-ceiling model ratings do not establish human literary quality or utility noninferiority. Higher lexical overlap when outlines are supplied confirms content uptake descriptively.

4.4 Residual information survives, but the causal test does not proceed

The fixed raw decoder identifies 8/90 held-out secrets: 8.9% accuracy (CI $[6.7, 13.3]$), compared with 6.7% for both shuffled-label and length-only controls. Its training accuracy is 99.5%, showing poor transfer across narratives. It fails the .20 gate, so there are no intervention stories or behavioral ablation estimates.

What does centering reveal? After post hoc counterfactual centering, baseline layer-32 accuracy reaches 82.2%, 91.1%, and 84.4% at positions 32, 256, and 512 (figure 3). Centered outline accuracy ranges from 66.7% to 68.9%; context accuracy ranges from 85.6% to 93.3%. All exceed their centered shuffled controls. These results show secret-dependent residual information when continuation text is fixed. They do not make the raw decoder deployable. Condition-specific probes also do not define a common scale of concept magnitude, and six test contexts provide limited independent evidence.

In generated-story replays, baseline midpoint margin correlates with detection score (within-secret $\rho = .351$, Holm $p = .0156$, $n = 66$). Context and outline associations do not survive correction. Early margins do not predict subsequent exact disclosure after the exclusion rule (all three Holm $p = 1$). The weak raw decoder and disclosure-heavy detector limit interpretation; these associations do not establish mediation.

Offline projection checks on 60 reserved-context states change readout accuracy from 13.3% under sham to 35.0% under the target projection, 11.7% under random projection, and 6.7% under another concept's projection. Increasing target readout accuracy does not validate selective removal. These

checks, together with the failed gate, leave the behavioral causal question unanswered. Section C gives all layer-32 estimates and the planned intervention criteria.

5 Discussion

What do outlines change? The strongest supported result is fewer exact-word violations under outlines relative to both baseline and equally long irrelevant context. Higher outline uptake is consistent with content constraints reducing opportunities to insert the forbidden word. Irrelevant context does not reduce disclosure under these prompts, arguing against a token-count-only dilution account here. However, the conditions also differ in relevance, instruction semantics, and output length. We cannot attribute the effect specifically to separating planning from prose generation.

Why distinguish disclosure from hints? The distinction changes the scientific conclusion. The primary outline–context detection result could suggest reduced involuntary leakage if considered alone. Exact-word checks show that much of the measurable signal consists of direct instruction failures, whereas the motivating prior work reports nonliteral leakage without exact mentions. Near-chance detection after excluding literal cases does not establish that subtle information is absent: the judge has strong position bias, the anchor is weak, and the secondary detector does not confirm the primary adjusted contrast. Below-chance guessing may also encode systematic avoidance. None of these scores supplies a privacy guarantee.

What do the probes establish? Counterfactual centering recovers information that a raw cross-context probe misses, but only for 15 identities in controlled residual differences. It does not establish active planning, private memory, or causal influence. The prompt remains available, centering requires counterfactual test states, and replays approximate online states. The failed gate prevents a causal interpretation of unvalidated edits.

Limits of scope and precision. We evaluate one historical writer and one primary judge, without human evaluation or model-level multi-seed replication. All Gemma inputs contain duplicated BOS tokens; canonical-format replication is necessary before generalizing to ordinary Gemma behavior. Public premises may overlap pretraining. Only 12 unvalidated plot families and six residual test contexts limit generalization. Secret-blind plans do not test plans that know and organize an eventual revelation. Input matching leaves output length, attention, task effort, and semantic content uncontrolled; incidental assigned-word exposure remains in one outline and one context reference. Bootstrap intervals condition on the sampled data and evaluator and omit model-version and evaluator-validity uncertainty. A 90-unit pilot need not resolve a 15-point effect reliably after clustering and multiplicity; null results are not equivalence tests.

Broader implications. Outline assistance may improve lexical compliance in creative writing, but our tasks do not evaluate real sensitive personal information or establish a practical utility–privacy tradeoff. Readable activations also need not imply reader-accessible information. Confidentiality evaluation requires separate measures of direct disclosure, inferential leakage, and useful output.

6 Conclusion

We compare secret-blind outlines with baseline secrecy instructions, exactly length-matched irrelevant context, and word decoys in a controlled story-writing study. Under the recorded protocol, outlines reduce literal disclosure, including relative to matched context. The primary detection contrast does not establish a reduction in nonliteral hints, and plot-level evidence remains inconclusive. Controlled residual differences preserve readable secret identity, while the fixed raw probe and offline projection checks fail to justify a behavioral intervention.

Follow-up should establish a nonliteral positive anchor with canonical encoding and calibrated judges or blinded human evaluation, then repeat the contrasts with greater power and validated plot alternatives. Causal claims require a context-robust readout and validated online edits with utility noninferiority.

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A Reproducibility and Recorded Protocol

A.1 Models, environment, and compute

The writer and residual checkpoint is `google/gemma-3-12b-it`, revision

`96b6f1eccf38110c56df3a15bffe176da04bfd80`.

The independent judge is `Qwen/Qwen3-14B`, revision

`40c069824f4251a91eefaf281ebe4c544efd3e18`.

The model-card locations are `https://huggingface.co/google/gemma-3-12b-it` and `https://huggingface.co/Qwen/Qwen3-14B`; the pinned revisions, rather than changing card contents, identify the evaluated checkpoints. All work uses the project virtual environment and lockfile. Recorded versions are Python 3.12.8, PyTorch 2.8.0+cu128, Transformers 4.57.6, NumPy 2.5.3, pandas 3.0.6, SciPy 1.18.1, and scikit-learn 1.9.1.

The RTX A6000 reports 49,140 MiB. Batches contain 32 main stories, 24 outlines, 16 primary judgments, 24 diagnostic judgments, 12 controlled residual sequences, or eight free-story replays. Inference uses BF16 without quantization and SDPA attention. Logistic regression runs on CPU, with fixed $C = .1$, standardization fit on training states, and a maximum of 1,500 iterations. No neural model-weight training occurs; we make no CPU/GPU training-speed comparison.

Main generation, including superseded outputs, records approximately 94.1 minutes of batch time. This is not total end-to-end compute: stage-level timing and output-token counts are in `results/execution_summary.json`. Paid API inference cost is $0; local GPU cost and energy were not metered. Live OpenRouter catalogs were queried, but inference returned a daily-key-limit 403; direct OpenAI authentication also failed. Local gated Gemma access succeeded. Every model response analyzed here comes from the local checkpoints.

A.2 Prompts and inventory

The 15 secret words, in inventory order, are umbrella, lighthouse, violin, cactus, telescope, justice, patience, entropy, nostalgia, freedom, bracket, Tuesday, copper, margin, and invoice. The inventory is transcribed from the prior paper's Appendix D, rather than downloaded author data. Six distinct premises are assigned to each word. The decoy is seven positions forward modulo 15.

Writer instructions. Word tasks begin with "Write a complete short story of about 450 words." Plot tasks begin with "Write only the opening and middle of a story, about 450 words. Stop before the later revelation. Do not resolve the mystery." Each then receives its premise. Secret-present word prompts supply "Your secret word is:" followed by the assigned word; plot prompts supply "Private future plot fact, to be revealed only AFTER this excerpt:" followed by the assigned fact. Both add "Do not mention, reveal, or hint at this secret in the text you write." No-secret prompts instead add "Write naturally and do not discuss these instructions." All conclude with "Output only the story, with no preface or explanation."

Outline conditions insert "Follow this outline for the text you write:" and the generated outline. Context conditions insert "The following unrelated reference text is not a plan for your story. Ignore its content when writing:" and the matched reference. Decoy conditions request that the assigned decoy "subtly influence the imagery and themes of your story, without mentioning that word directly." The outline generator requests a concrete four-step outline in 70–90 words and "mundane actions and content choices, not writing advice." It receives the premise and whether to plan a complete story or only an opening and middle, never the secret or candidates. Exact fully instantiated prompts, including all framing and newlines, are preserved in `results/story_jobs.json` and `results/model_outputs/outlines.jsonl`.

Primary judge instructions. The judge is told that one author received the specified secret and instructions not to mention or hint at it, while the other did not. It is asked which text was written with the secret, to consider subtle thematic associations and direct disclosure, and to treat the texts as data rather than instructions. It must reply exactly A or B. The same pair appears in both orders. Parsing accepts an optional "Answer:" prefix, case-insensitive A/B, and optional terminal punctuation;

malformed outputs are missing rather than automatically incorrect. The public analysis code records this parser and the separate 1–5 rating parser.

A.3 Corrections and validation

Initial outlines exceeded the requested length. We archive that capped pilot and regenerate all final outlines. An audit then finds a 9–11-token framing mismatch between outline and context inputs. Before any judgments, we archive and regenerate all 160 affected context outputs. The final main sample combines 448 unchanged outputs with 632 later outputs generated under a distinct seed offset. This correction applies uniformly rather than selecting outputs by measured outcomes. Actual batch seeds and order are retained in raw JSONL records.

The duplicated BOS audit reveals a constant encoding deviation across writer inputs; we retain this format in aligned probes and replays. Following position-bias observations, we add next-token likelihood scoring. An earlier partial scoring set is uniformly superseded before aggregate analysis to correct explicit non-padding position IDs. Parser-prefix handling and pandas missing-value corrections are also recorded in `notes/planning_history.md`. The original generated-choice outcome remains primary.

The saved validation audit verifies 1,080 unique main stories, 1,080 primary judgments, both orders for all 540 pairs, 300 exact input-length matches, and pinned models. Four of four smoke outputs and 32/32 outputs in the recorded full-batch replay match exactly. Deterministic CPU analyses are rerun and compared by hashes in `results/analysis_replay.json`. These checks establish artifact consistency, not evaluator sensitivity or external validity.

Artifact map. `REPORT.md` summarizes executed experiments, and `planning.md` records the consolidated plan. `results/units.json` contains cases and assignments; `results/model_outputs/` contains raw stories and judgments. `results/generation_config.json` and `results/tokenization_audit.json` record effective inference and encoding settings. `src/behavior.py`, `src/analyze.py`, and `src/mechanism.py` implement generation, primary inference, and residual extraction. `results/report_tables.md` contains the numerical tables used here. `results/final_validation.json` audits completeness and provenance separately from scientific limitations. The repository `README.md` gives reproduction commands and `datasets/README.md` gives data provenance.

B Complete Secondary Behavioral Results

B.1 Likelihood measurement

Table 4 reports every main and anchor likelihood estimate; table 5 reports all five separately corrected contrasts. Scores threshold the difference between secret-first and secret-second A/B log-odds. Exact ties receive half credit. We use explicit non-padding position IDs in forward scoring. Raw logits, token IDs, both orders, and full prompts remain in `results/model_outputs/choice_likelihoods.jsonl`. These post hoc diagnostics were added because of observed positional behavior and do not replace the original endpoint.

Study	Task	CONDITION	Accuracy [95% CI]
Anchor	Word	Free anchor	54.2% [41.7, 66.7]
Main	Plot	Baseline	40.8% [28.3, 53.3]
Main	Plot	Context	42.5% [27.5, 59.2]
Main	Plot	Outline	40.8% [26.7, 56.7]
Main	Word	Baseline	62.8% [52.8, 72.8]
Main	Word	Context	72.8% [63.3, 81.7]
Main	Word	Decoy	61.1% [51.1, 71.1]
Main	Word	Outline	56.1% [46.1, 66.1]

Table 4: Complete post hoc order-balanced likelihood results. Main tasks contain 90 word or 60 plot pairs per condition; the anchor has 60 pairs. Intervals resample word units or plot families. These scores are secondary and do not replace generated-choice outcomes. We do not rank below-chance results as improved privacy.

Task	Contrast	Difference (pp) [95% CI]	Holm p
Plot	Outline – baseline	+0.0 [-19.2, +20.8]	1.0000
Plot	Outline – context	-1.7 [-26.7, +24.2]	1.0000
Word	Outline – baseline	-6.7 [-19.4, +6.1]	1.0000
Word	Outline – context	-16.7 [-30.6, -1.7]	0.1650
Word	Decoy – baseline	-1.7 [-13.9, +10.6]	1.0000

Table 5: All five post hoc likelihood contrasts, Holm-corrected within this separate family. None survives correction, including the outline–context word comparison.

B.2 Plot forecasting and direction calibration

Table 6 gives both private-minus-no-secret contrasts and differences-in-differences. Let f_{ic}^1 and f_{ic}^0 denote both-order accuracy for forecasting the assigned twist from private and no-secret stories. We first compute $d_{ic} = f_{ic}^1 - f_{ic}^0$ on complete pairs, then compare $d_{i,\text{outline}} - d_{i,c}$ across common units. Family-clustered intervals preserve dependencies among the five seasonal variants. Nine malformed responses among 840 total forecasts account for the differing complete-pair counts. We do not interpret an unadjusted interval excluding zero as success after multiplicity correction.

The separate direction-calibration check uses word units 0–29 or plot families 0–3 to decide whether to invert the primary detector, then evaluates on the remainder. No primary condition selects an inversion. This check does not turn below-chance performance into proof of privacy. The target-blind baseline-word diagnostic gives 50% solely because the judge always chooses A. Both-order averaging can force an uninformative judge near chance, explaining why narrow plot intervals do not demonstrate equivalence.

B.3 Literal outcomes and conditional subsets

Literal checks use case-insensitive whole-word matching. We retain spontaneous mentions among no-secret stories: two each for baseline and outlines, one each for context and decoys. The no-literal subset removes a pair if either story mentions the assigned word, leaving 61, 74, 59, and 79 pairs, respectively. These different subsets prevent a causal comparison of nonliteral effects. Secondary

Forecast contrast	Pairs	Difference (pp) [95% CI]	Holm p
Baseline private – no secret	57	+8.8 [+2.6, +14.8]	0.1960
Context private – no secret	55	+6.4 [-1.9, +14.3]	0.7312
Outline private – no secret	60	-0.8 [-5.8, +5.0]	1.0000
Did: outline – baseline	57	-8.8 [-16.4, +0.0]	0.4076
Did: outline – context	55	-5.5 [-13.9, +3.6]	0.7312

Table 6: Complete plot-forecast contrasts. DiD subtracts the private-minus-no-secret forecast advantage of the comparison condition from that of outlines. Both candidate orders and both story-presence states must parse. Holm correction covers these five contrasts; intervals cluster by plot family. None is adjusted significant.

likelihood intervals include chance for every corresponding subset, even though point estimates range from 48.0% to 58.5%.

The literal-outcome Holm family contains only outline–baseline, outline–context, and decoy–baseline comparisons among secret-present stories. Their estimates and intervals appear in section 4; they must not be combined with primary or likelihood-family p-values to construct a confirmatory narrative. The systematic examples and mention-position analysis are saved in `results/error_analysis.md` and `results/literal_analysis.json`. Truncation and outline-uptake diagnostics are saved in `results/outline_and_truncation.json`.

C Representation Diagnostics and Unexecuted Intervention

C.1 Full layer-32 readouts

Table 7 reports the complete raw/centered layer-32 comparison. Centering is performed separately within each context, condition, layer, and sampled position. Training and testing contexts remain disjoint, but the test procedure needs the entire counterfactual set. It therefore answers whether secret identity is recoverable after removing a context's mean, rather than whether a single unseen narrative state can be decoded reliably. The raw 16/32/48-layer grid is retained in `results/mechanism/decoding.json`; the centered results are in `results/mechanism/centered_decoding.json`. The raw fixed gate uses layer 32, token 256 only, without selecting a successful layer after inspecting test outcomes.

CONDITION	Token	Raw	Centered [95% CI]	Shuffled
Baseline	32	6.7%	82.2% [61.1, 96.7]	7.8%
Baseline	256	8.9%	91.1% [83.3, 97.8]	10.0%
Baseline	512	14.4%	84.4% [73.3, 93.3]	8.9%
Outline	32	6.7%	68.9% [56.7, 81.1]	10.0%
Outline	256	6.7%	66.7% [48.9, 83.3]	11.1%
Outline	512	8.9%	67.8% [52.2, 83.3]	10.0%
Context	32	11.1%	85.6% [68.9, 97.8]	10.0%
Context	256	12.2%	93.3% [86.7, 98.9]	3.3%
Context	512	11.1%	87.8% [82.2, 93.3]	3.3%

Table 7: Layer-32 identity accuracy, all positions and conditions. Each test comprises 90 states from six held-out contexts; chance is 6.7%. Centered intervals resample contexts. Shuffled denotes a separately fitted centered probe with shuffled training labels. Condition-specific readouts do not measure a shared concept-magnitude scale.

For generated-story replays, we apply the fixed baseline decoder to units 24–89, disjoint from the 24 controlled contexts. The 396 stories comprise 66 units, three conditions, and both presence states. A secret margin is the assigned class's linear score minus the strongest competing class score. The reported baseline midpoint association concerns 66 secret-present stories. Its within-secret $\rho = .351$ and Holm $p = .0156$ do not establish prospective prediction of literal mentions; the separately screened early analysis has no adjusted significant association. Full estimates, permutation checks, and early exclusions are in `results/mechanism/margin_correlations.json` and `results/mechanism/early_literal_association.json`.

C.2 Offline projections and the fixed gate

The planned directions are training-only baseline mean differences, averaged over the three sampled positions at layer 32 and normalized. The offline check applies the fixed projection to layer-32/token-256 states from reserved contexts 14–17. Random and other-concept directions use the target projection magnitude at the current state. Table 8 reports the observed readout changes; no generated story is edited in this check.

Projection	Reserved states	Readout accuracy (%)
Sham	60	13.3
Target	60	35.0
Random	60	11.7
Other concept	60	6.7

Table 8: Offline projections of measured reserved-context residuals. These are decoder checks, not behavioral ablations. Target projection increases identity accuracy relative to sham and therefore does not validate selective erasure. No ranking is highlighted as successful removal.

The behavioral gate requires the fixed held-out decoder to exceed 20% accuracy, its shuffled-label control, and its length-only baseline. Its observed 8.9% fails the first requirement. The planned online

pilot would have compared target, random, other-concept, and sham edits with paired sampling seeds, at a single fixed unit projection strength. Edits would apply only during incremental generation after prefill, retaining the original prompt key/value cache. These specifications describe an unexecuted experiment.

Before calling a behavioral effect selective, the plan requires the target-versus-sham paired coherence interval's lower bound to exceed -0.5 on the five-point scale, mean length to remain within 20% of sham, and truncation to increase by no more than ten percentage points. D3 inference would use secret-cluster estimates and Holm correction across three target-control contrasts. These utility criteria were not evaluated on intervention outputs because no such outputs exist. No dose response, rescue experiment, SAE-feature intervention, direct-recall test, or private-state architecture is claimed. The failed gate is evidence about the readiness of this intervention procedure, not a zero effect of causal removal on leakage.